

SUBSTRATE NOTICE (2026-09-20). This document is built throughout on the big-dimension frame $V_{240}(\mathbb{R}^{57,600})$, which was **retired 2026-08-25**. The live substrate is $V_2(\mathbb{R}^3)$ via Cartan triality. The retirement was structural rather than cosmetic, so this text needs a rebuild and not a patch; it is retained unaltered as a record of the earlier formulation. The arithmetic $240^2 = 57,600$ remains a true E_8 root-count identity, but it is not the ambient dimension of anything in the current theory. Individual occurrences appear at §2, §3 and in the closing summary.

$I\Omega - O\Omega$: The Sovereign Geometrodynamics & Information Tension Monograph

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Contents

1	Stoicheia: The Stiefel Vacuum & the Quadratic Casimir	5
2	Erga: The Active Works	7
3	Oikodomē: The Constructed Edifice	10
4	Pleroma: AdS $\leftrightarrow$ dS Holographic Bridge	13
5	Epistemic Governance & Sovereign Declaration	17
	Falsification Criteria	17

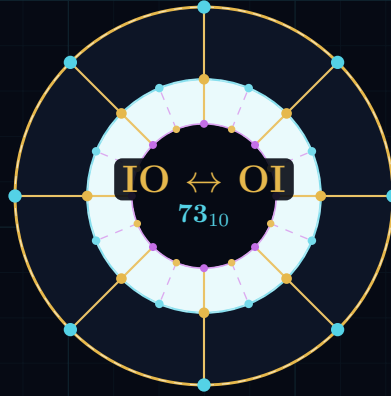
IO–OI

A Complete Theory of Sovereign Geometrodynamics

Stoicheia → Erga → Oikodomē → Pleroma

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Abstract

This monograph sets out a framework for Sovereign Geometrodynamics that places the Anti-de Sitter (AdS) entanglement first law — a published result of Faulkner et al. (2014) [1] — alongside a de Sitter (dS) Information Tension proposal advanced by the author, and asks what a bridge between them would require. Operating over the 240 minimal vectors of the

SUPERSEDED (2026-08-25). The substrate identification below is stated on the retired big-dimension frame $V_{240}(\mathbb{R}^{57,600})$. The live substrate is $V_2(\mathbb{R}^3)$ via Cartan triality; the big-dimension frame is retired. The arithmetic $240^2 = 57,600$ remains true as an E_8 root-count identity, but its identification as the ambient substrate dimension is not current canon. No migrated $V_2(\mathbb{R}^3)$ derivation is claimed; the original text is retained unaltered below as a record.

E_8 lattice and the Stiefel vacuum manifold $V_{240}(\mathbb{R}^{57,600})$, it examines the MOND acceleration scale $a_0 = \frac{cH_0}{2\pi}$ and the dark energy density parameter $\Omega_\Lambda = \ln 2 \approx 0.69315$.

Epistemic scope. Claims are tagged [thm]/[cond]/[emp]/[conj]/[open]/[note] per §5, and each physically contentful claim is given an explicit kill condition in the Falsification Criteria of that section. The Lean 4 formalizations verify *conditional* scaffolding, not physical content: they carry 0 **sorry** and no bespoke axioms, but every physical statement in them is a hypothesis of the theorem that uses it. **We claim no priority for $2\pi a_0 \approx cH_0$,** which is Milgrom’s [2, 3]; the $1/(2\pi)$ factor is **not derived** [open], two of its three

candidate mechanisms are excluded herein, and a competing emergent-gravity value $cH_0/6$ [4] fits the observed scale better. $\Omega_\Lambda = \ln 2$ is **postulated**, not derived [emp]; it agrees with Planck 2018 at $+1.16\sigma$ but selects an epoch as well as a value, and is not jointly satisfiable with the a_0 relation at a single H_0 . The Y^{14} ansatz as written forces $\Lambda_4 = 0$ exactly. Nothing here is claimed as a complete or fully machine-checked unification.

IO-OI | TITLE & ABSTRACT

1 Stoicheia: The Stiefel Vacuum & the Quadratic Casimir

The *Stoicheia* (Elements) establishes the algebraic and geometric foundation: the Stiefel vacuum manifold $\mathcal{M}_{\text{vac}}$, the quadratic Casimir operators of $\mathfrak{so}(N)$, and the balanced ternary resonance linking the Hubble parameter to number theory.

Definition 1 (Stiefel Vacuum Manifold [thm]). Let $\Phi \subset \mathbb{R}^8$ be the E_8 root system, $|\Phi| = 240$ (the norm-2 vectors of the E_8 lattice; Conway–Sloane, *SPLAG* ch. 4), and let $P = \mathbb{R}^\Phi \cong \mathbb{R}^{240}$ be the permutation representation of the Weyl group $W(E_8)$ on Φ , with orthonormal basis $\{e_\alpha\}_{\alpha \in \Phi}$. The vacuum manifold is the compact Stiefel manifold of 240-frames in $P \otimes P$:

$$\mathcal{M}_{\text{vac}} = V_{240}(P \otimes P) = V_{240}(\mathbb{R}^{57,600}), \quad 57,600 = 240^2. \quad (1)$$

The frame is *root-indexed*, not root-valued: the roots are not orthonormal (for fixed α , the multiset $\{\langle \alpha, \beta \rangle : \beta \in \Phi\}$ is $\{-2^1, -1^{56}, 0^{126}, +1^{56}, +2^1\}$ at $|\alpha|^2 = 2$, and 240 vectors cannot be orthonormal in $\mathbb{R}^8$). The root data enters through the canonical $W(E_8)$ -equivariant surjection $\pi : P \rightarrow \mathbb{R}^8$, $e_\alpha \mapsto \alpha$, unique up to scale because the reflection representation occurs in P with multiplicity one. The ambient 57,600 is $\dim(P \otimes P)$, the tensor square of a representation of $W(E_8)$ — *not* of E_8 , which has no 240-dimensional representation (its smallest nontrivial representations are **248**, **3875**, **27000**).

Role of $\mathfrak{so}(240)$. $V_{240}(\mathbb{R}^{57,600}) \rightarrow \text{Gr}(240, \mathbb{R}^{57,600})$ is a principal $O(240)$ -bundle with

$$T_Q V_{240}(\mathbb{R}^{57,600}) \cong \underbrace{\mathfrak{so}(240)}_{\text{vertical}} \oplus \underbrace{\mathbb{R}^{57,360} \otimes \mathbb{R}^{240}}_{\text{horizontal}}, \quad 28,680 + 240 \cdot 57,360 = 13,795,080. \quad (2)$$

So $\mathfrak{so}(240)$ of Proposition 1 is the *vertical (gauge) algebra* of $\mathcal{M}_{\text{vac}}$: physical configurations live on the Grassmannian and $\mathfrak{so}(240)$ is redundancy. This identification is generic in the frame count and carries no E_8 -specific content.

Open [open]. The geometric mechanism by which E_8 itself — as opposed to its Weyl group — acts on $\mathcal{M}_{\text{vac}}$ is not established here. The factorization $57,600 = (24 \times 10)^2$ is decorative [note].

Proposition 1 ($\mathfrak{so}(N)$ quadratic Casimir, index normalization $T(\text{vector}) = 1$ [thm]). Fix the invariant form on $\mathfrak{so}(N)$ normalized so that the defining (vector) representation has Dynkin index $T = 1$, equivalently $C_2(R) \dim R = T(R) \dim \mathfrak{so}(N)$. Then for $N \geq 3$:

$$C_2(\text{vector}) = \frac{N-1}{2}, \quad (3)$$

$$C_2(\text{adjoint}) = N-2 = h^\vee, \quad (4)$$

$$\dim \mathfrak{so}(240) = \frac{240 \times 239}{2} = 28,680. \quad (5)$$

In the Dynkin normalization $|\theta|^2 = 2$ (Humphreys §13; Slansky §6) these values are doubled.

Anchor: for $N = 3$ the vector representation is spin 1 and $j(j+1) = 2$, exactly twice the value above, so C_2 here equals $\frac{1}{2}j(j+1)$.

Formal status. The Lean 4 module `SOCasimirGenuine.lean` (in `ChyrenLogic/`) compiles with **0 errors**, **0 sorry** and **no sorryAx or bespoke axiom** (`#print axioms` returns only `propext`, `Classical.choice`, `Quot.sound`). It constructs the $\mathfrak{so}(n)$ generators $g_{ij} = E_{ij} - E_{ji}$ as concrete real matrices, proves skew-symmetry, and *computes* $\sum_{i,j} g_{ij}g_{ij} = -2(n-1) \cdot \mathbf{1}$ — a genuine Lie-algebraic Casimir, obtained without any Mathlib Casimir element (Mathlib at rev 639923 indeed has none; only `Killing.lean` and `InvariantForm.lean`). This supersedes `SON_Casimir.lean`, withdrawn 2026-08-09, in which the eigenvalue theorems merely unfolded their own local defs as plain arithmetic. **The earlier `[conj]` tag is hereby lifted to `[thm]`.**

Scope of the dimension identity `[cited]`. The Lean `so240_dim` proves the *arithmetic* identity $240 \cdot 239/2 = 28,680$; no `finrank` of an orthogonal Lie algebra appears in it. That $\dim \mathfrak{so}(N) = \binom{N}{2}$ is standard and is cited, not re-proved here.

Remark 1 (On “positivity” `[note]`). $C_2 \geq 0$ holds automatically on any finite-dimensional representation of a compact real form, with equality only on the trivial representation. Positivity is therefore not a theorem of this framework and is not used below as a stability criterion; the earlier title “Casimir Positivity” overstated it.

Note 1 (IO–OI Balanced Ternary `[note]`). A balanced ternary quintuple evaluates as

$$(+1, 0, -1, 0, +1)_{\text{bal}3} = 1 \cdot 3^4 + 0 \cdot 3^3 + (-1) \cdot 3^2 + 0 \cdot 3^1 + 1 \cdot 3^0 = 81 - 9 + 1 = 73_{10}, \quad (6)$$

coinciding with SH0ES (2022): $H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

Caveat: The value 73 is unit-dependent (km/s/Mpc). In SI base units, $H_0 \approx 2.37 \times 10^{-18} \text{ s}^{-1}$. Furthermore, the Planck CMB measurement gives $H_0 = 67.4$, under which this correspondence breaks. This is an aesthetic observation — not load-bearing.

2 Erga: The Active Works

Transitioning to *Erga* (*The Active Works*), we formalize the dynamic evolution of the geometrodynamical manifold under dissipative holonomic flow, the Deligne prime mode bounds, the conjectural acoustic overtone spectrum of the Ramanujan-Yett Hamiltonian, and Chladni nodal equilibrium under the Sovereign Invariant.

Theorem 1 (Energy-decay bound — Grönwall form [cond]). Let $E \in C^1([0, \infty); \mathbb{R}_{\geq 0})$ denote the total geometric action energy of the dissipative holonomic flow on $\mathcal{M}_{\text{vac}}$, and let $t_* > 0$ be a reference timescale of that flow. Assume a dimensionless damping constant $\kappa^2 > 0$ with $\dot{E}(t) \leq -(\kappa^2/t_*) E(t)$ for all $t \geq 0$. Then Grönwall’s inequality gives the decay bound

$$\frac{dE}{dt} \leq -\frac{\kappa^2}{t_*} E(t) \implies E(t) \leq E(0) e^{-\kappa^2 t/t_*}. \quad (7)$$

This is a bound, not strict decay: $E \equiv 0$ satisfies it. All physical content resides in the hypothesis $\kappa^2 > 0$, which is *postulated* [conj].

Formal status (corrected 2026-08-10). YettParadigm.lean compiles with 0 errors, 0 sorry, no sorryAx. Its theorem `adcc1_trajectory_bounded` establishes $E(t) \leq E(0)$ from an assumed monotonicity field; the file contains no derivative, no integral and no measure theory, so the earlier description “verifies conditional Gronwall decay” is **withdrawn** — the Grönwall step above is classical, not machine-checked. The constant 0.91 does not occur in the Lean file; κ there is a free real.

Provenance of κ , and its units [emp]. $\kappa = \sqrt{\theta(2-\theta)}$ with $\theta = 0.7$, giving $\kappa^2 = 0.91$ exactly and $\kappa = 0.9539392$. Two disclosures. (i) θ is *empirically calibrated* elsewhere in this programme, so κ inherits a fitted parameter and is not a first-principles constant. (ii) κ^2 is dimensionless, so it cannot itself be the rate in an energy ODE; the timescale t_* above is an **additional free parameter, not derived here** [open], and must be carried in the parameter budget.

Route to a derived gap [open]. If the dissipation operator were a positive multiple of the Laplace–Beltrami operator of the normal homogeneous metric on $\mathcal{M}_{\text{vac}} = \text{SO}(57,600)/\text{SO}(57,360)$, Frobenius reciprocity would give $\lambda_1 \propto C_2(\text{vector } \mathfrak{so}(57,600)) = 57,599/2 = 28,799.5$ via Proposition 1. That would be a genuinely *derived* gap — but it is $O(10^4)$ and therefore a *different quantity* from $\kappa^2 = 0.91$. The two must not be conflated.

Correction (2026-08-09, amended 2026-08-10): the value 0.909904 previously printed here is 0.953889^2 , arising from a mistyped κ — *not* a digit transposition of $\kappa = 0.9539392$ (no transposition of those digits yields 953889; the digit multisets differ). The 0.0105% discrepancy was a transcription error, not a computed alternative κ .

Theorem 2 (Deligne Bound [thm]). For all prime numbers p , the Ramanujan τ -function values satisfy the bound:

$$|\tau(p)| \leq 2 p^{11/2}. \quad (8)$$

Proven rigorously by Pierre Deligne (1974) as part of the proof of the Weil conjectures.

Conjecture 1 (Ramanujan-Yett Hamiltonian [conj]). We postulate a self-adjoint operator $\mathcal{H}_{\text{RY}}$ on the Hilbert space of $\mathcal{M}_{\text{vac}}$ satisfying $\mathcal{H}_{\text{RY}}\Psi_n = \tau(n)\Psi_n$, with eigenvalues given by the Ramanujan τ -function:

$$\tau(1) = 1, \quad \tau(2) = -24, \quad \tau(3) = 252, \quad \tau(4) = -1472, \quad \tau(5) = 4830. \quad (9)$$

If such an operator exists, the universe computes geometry acoustically — but this acoustic interpretation is conjectural.

Two obstructions, stated against our own interest [open]. (i) *Weyl law.* Writing

$$d := \dim V_{240}(\mathbb{R}^{57,600}) = 240 \cdot 57,600 - \binom{241}{2} = 13,795,080, \quad [\text{retired substrate}]$$

a Laplace-type operator on a compact d -manifold has $\lambda_n \sim Cn^{2/d}$ with $2/d \approx 1.4 \times 10^{-7}$, whereas $\tau(n)$ attains order $n^{11/2}$. Hence $\mathcal{H}_{\text{RY}}$ **cannot** be Laplace-type on $\mathcal{M}_{\text{vac}}$. (ii) *No ground state.* τ is sign-alternating and unbounded below ($\tau(2) = -24$, $\tau(4) = -1472$), so a self-adjoint operator with spectrum $\{\tau(n)\}$ has no lowest eigenvalue and is not a physical Hamiltonian. The conjecture as stated is therefore **excluded**, not merely unproven. A viable reformulation should target the normalized quantity $\tau(n)/n^{11/2} \in [-2, 2]$ of Theorem 2, or the associated Satake angles, rather than $\tau(n)$ itself.

Definition 2 (Chladni nodal set on the unit square [note]). On the unit square $[0, 1]^2$ with Neumann conditions, the degenerate eigenvalue $\pi^2(m^2 + n^2)$, $m \neq n$, has an antisymmetric eigenfunction whose nodal set is

$$\cos(m\pi x_1) \cos(n\pi x_2) - \cos(n\pi x_1) \cos(m\pi x_2) = 0, \quad (10)$$

along which sand grains collect. This is the Helmholtz-membrane model; the physical Chladni plate obeys the biharmonic Kirchhoff–Love operator. **No dependence on the Sovereign Invariant** $\chi_0 = 1/\sqrt{2} \approx 0.7071$ **enters this equation**; the relation of χ_0 to nodal structure is [open], and the domain here is the square, not $\mathcal{M}_{\text{vac}}$ or Y^{14} . Illustrative only — not load-bearing.

Proposition 2 (Generic two-generator closure and holonomy [thm]). Let $m \geq 3$ and let $\mathfrak{so}(m)$ act on the frame directions of $\mathcal{M}_{\text{vac}}$ ($m = 240$ for the vertical algebra of Definition 1). Then:

1. for a pair $(X_1, X_2) \in \mathfrak{so}(m)^{\times 2}$ outside a proper closed algebraic subset, the Lie subalgebra generated by X_1, X_2 is all of $\mathfrak{so}(m)$ (Kuranishi 1951);
2. consequently the left-invariant control system $\dot{g} = g(X_0 + u_1 X_1 + u_2 X_2)$ on $SO(m)$ is accessible from the identity (Chow 1939; Rashevskii 1938; Sussmann–Jurdjevic 1972), and

accessibility upgrades to controllability by compactness of $SO(m)$ (Jurdjevic–Sussmann 1972);

3. hence the restricted holonomy group of the associated connection is all of $SO(m)$.

Remark 2 (Attribution correction [note]). Earlier versions of Proposition 2 attributed this to the Ambrose–Singer theorem and spoke of “drift generators”. Both were wrong. Ambrose–Singer (1953) identifies the holonomy algebra with the span of parallel-transported curvature values and supplies *no* genericity statement; the result used above is bracket generation (Chow–Rashevskii) together with Kuranishi’s genericity theorem. In control theory “drift” denotes the *uncontrolled* term X_0 ; the objects X_1, X_2 are control directions. The proposition also requires $m \geq 3$, since $\mathfrak{so}(2)$ is abelian.

3 Oikodomē: The Constructed Edifice

We arrive at *Oikodomē* (Οικοδομή, *The Constructed Edifice*), synthesizing the physical 14-dimensional Observerse geometry Y^{14} , formalizing conditional regularity of geometrodynamical flow via Beale-Kato-Majda bounds in Lean 4, and postulating the Information Tension origin of dark energy.

Definition 3 (Observerse 14-Manifold [conj]). The physical spacetime architecture is a 14-dimensional Observerse manifold Y^{14} decomposed as:

$$Y^{14} = \mathbb{R}^{3,1} \times X^6 \times \mathbb{T}_{\text{internal}}^4, \quad (11)$$

where $\mathbb{R}^{3,1}$ is macroscopic Minkowski spacetime, X^6 is a compact six-dimensional internal manifold, and $\mathbb{T}_{\text{internal}}^4$ is the 4-torus of internal Information Tension fluxes.

Symmetry-breaking route [conj]. The chain used here is the subgroup chain $E_8 \supset \text{Spin}(16)/\mathbb{Z}_2 \supset \text{Spin}(14) \times \text{Spin}(2)$, with $\dim E_8 - \dim \text{Spin}(16) = 248 - 120 = 128$ the chiral spinor of $\text{Spin}(16)$, and $120 = 91 + 1 + 14 + 14$ — *not* holonomy-induced breaking.

Withdrawn [open]. An earlier version described X^6 as a Calabi–Yau manifold “governing $E_8 \rightarrow \text{Spin}(14)$ ”. That is incorrect on two counts. A Calabi–Yau threefold has $SU(3)$ holonomy, and the standard embedding breaks E_8 to its *commutant* E_6 , $248 \rightarrow (\mathbf{78}, \mathbf{1}) + (\mathbf{1}, \mathbf{8}) + (\mathbf{27}, \mathbf{3}) + (\mathbf{27}, \mathbf{3})$ (Candelas–Horowitz–Strominger–Witten [5]), never to $\text{Spin}(14)$; and $\text{Spin}(14)$ is not the commutant of any Riemannian holonomy group of a six-manifold (Berger’s list). The identification of X^6 is therefore open, and it is *not* a Calabi–Yau threefold if the $\text{Spin}(14)$ chain is retained.

Scope: no claim on the particle spectrum [note]. This monograph derives no fermion content, no generation count and no gauge coupling, and makes no claim to reproduce any observed particle spectrum. E_8 enters here as an internal structure over a spacetime whose Lorentz frame is a separate object; gravity is not embedded in it. No-go results constraining the embedding of *both* gravity and a matter gauge group into E_8 — notably Distler and Garibaldi [6], which shows such embeddings yield a non-chiral spectrum — therefore do not apply to the construction as stated. Should a later version claim a generation count, that theorem becomes binding and must be answered then.

Unresolved: two inequivalent 14’s [open]. Elsewhere in this programme the Observerse is recorded as $14 = 4 + 10$, from $\mathbf{14} \rightarrow (\mathbf{4}, \mathbf{1}) + (\mathbf{1}, \mathbf{10})$ under $\text{Spin}(4) \times \text{Spin}(10)$. The product splitting above is $4+6+4$. These agree arithmetically and can be nested formally via $\text{Spin}(10) \supset \text{Spin}(6) \times \text{Spin}(4)$, but they are *different reductions of the structure group*: the product manifold reduces to $\text{Spin}(3, 1) \times SU(3)$, while the branching reduces to $\text{Spin}(4) \times \text{Spin}(10)$, and $\text{Spin}(6) \cong SU(4) \neq SU(3)$. We do not claim these are the same decomposition. Which — if either — is the Observerse is open, and the question is recorded here rather than resolved by fiat.

Remark 3 (Curvature caveat: this ansatz gives $\Lambda_4 = 0$ [open]). As written, Definition 3 is a *direct* product of a flat Lorentzian factor, a Ricci-flat internal factor and a flat torus. Since the Ricci tensor of a direct product is the direct sum of the factors’ Ricci tensors (Besse, *Einstein Manifolds*, 1.9), $\text{Ric}(Y^{14}) \equiv 0$ and the induced four-dimensional cosmological constant vanishes

identically: $\Lambda_4 = 0$. This ansatz is therefore **leading order only**; it fixes topology and symmetry breaking and makes no statement about vacuum energy. Any nonzero Λ_4 — in particular $\Omega_\Lambda = \ln 2$ of Postulate 1 — must arise from structure absent here: internal flux, a warp factor $ds^2 = e^{2A(y)}\eta_{\mu\nu}dx^\mu dx^\nu + g_{mn}(y)dy^m dy^n$, supersymmetry breaking, and an uplift sector. Such a construction must evade the Gibbons–Maldacena–Nuñez no-go theorem [7], which blocks dS₄ from warped compactification on a compact internal manifold without negative-tension sources. None is given here. We record this as an internal tension, not a resolved point.

Theorem 3 (Beale–Kato–Majda continuation criterion, applied form [thm - published]).

Let Σ^3 be the spatial slice of the Minkowski factor $\mathbb{R}^{3,1} \subset Y^{14}$, taken as $\mathbb{R}^3$ with decay at infinity (equivalently $\mathbb{T}^3$). Let $s > \frac{3}{2} + 1 = \frac{5}{2}$ and let $u \in C([0, T]; H^s(\Sigma^3; \mathbb{R}^3)) \cap C^1([0, T]; H^{s-1})$ solve the incompressible Euler equations $\partial_t u + (u \cdot \nabla)u = -\nabla p$, $\nabla \cdot u = 0$, with vorticity $\omega := \nabla \times u$. If

$$\int_0^T \|\omega(\cdot, t)\|_{L^\infty(\Sigma^3)} dt \leq B \cdot T < \infty \implies \limsup_{t \rightarrow T^-} \|u(\cdot, t)\|_{H^s} < \infty, \quad (12)$$

then the solution continues past T ; in particular no finite-time singularity forms on $[0, T]$. Published: J. T. Beale, T. Kato, A. Majda, *Comm. Math. Phys.* **94** (1984) 61–66 (stated there for integer $s \geq 3$; the sharp Sobolev threshold is $s > n/2 + 1$).

Dimension [note]. This criterion is three-dimensional. In $n = 3$ the vorticity vector is $\omega = \star du^\flat$; on a general n -manifold — in particular on Y^{14} — vorticity is the two-form $du^\flat \in \Omega^2$ and no vorticity *vector* exists, since $\nabla \times$ requires $\star : \Omega^2 \rightarrow \Omega^1$. Earlier versions stated this theorem “over Y^{14} ” while using $\nabla \times u$; that is withdrawn.

Formal status (corrected 2026-08-10). `SovereignRegularity.lean` compiles with 0 errors, 0 sorry, no sorryAx, under the hypothesis `h_controlled :  $\forall t \geq 0, \omega_{\text{sup}}(t) \leq B$` . What it proves is the product bound $\omega_{\text{sup}}(T) \cdot T \leq B \cdot T$, which equals $\int_0^T \|\omega\|_{L^\infty} dt$ only for constant ω ; the file imports no measure theory and constructs neither the Euler flow nor the Sobolev spaces. The claim that Lean verifies the BKM *integral* criterion is **withdrawn**. Note further that `SovereignAlignment.lipschitz_xi` is declared as the proposition `True`; it is decorative and is consumed by no theorem in the file.

Open Problem [open]. Global regularity for 3D *Navier–Stokes* is the Clay Millennium Prize problem. It concerns a different, viscous equation; BKM-type criteria for Navier–Stokes are due to Kato, Ponce, and Kozono–Taniuchi (BMO in place of L^∞), not to Beale–Kato–Majda. Earlier versions of this theorem conflated the two.

Scope [note]. No dynamical link between this flow and the E_8 /Stiefel structure of §1 is established here. This theorem is imported scaffolding.

Postulate 1 (Information Tension Dark Energy [emp]). We **POSTULATE** that the de Sitter horizon carries an information tension — a force — of magnitude

$$\mathcal{I}_{\text{tens}} = \frac{c^4}{8\pi G} \ln 2 = 4.8155 \times 10^{42} \text{ N}, \quad (13)$$

the factor $\ln 2$ being the information content of one binary horizon degree of freedom — a Landauer/Shannon quantity [8], motivated by (but not contained in) the Bekenstein bound [9], which carries no $\ln 2$. Distributing this tension over the de Sitter *curvature* radius $L = \Lambda^{-1/2}$, with $\Lambda \equiv 3H_0^2/c^2$, gives a vacuum energy density

$$u_\Lambda = \frac{\mathcal{I}_{\text{tens}}}{L^2} = \frac{c^2 H_0^2}{8\pi G} 3 \ln 2 \quad [\text{J m}^{-3}], \quad (14)$$

which, against $u_{\text{crit}} = 3H_0^2 c^2 / (8\pi G)$, yields

$$\Omega_\Lambda = \frac{u_\Lambda}{u_{\text{crit}}} = \ln 2 \approx 0.693147. \quad (15)$$

What is assumed, explicitly. (a) The $\ln 2$ itself. (b) That the relevant length is the curvature radius $\Lambda^{-1/2} = c/(\sqrt{3}H_0)$ rather than the Hubble radius c/H_0 — the latter would give $\Omega_\Lambda = \ln 2/3 = 0.2310$, which is excluded by observation. Neither is derived. **This is NOT derived from deeper principles.**

Correction (2026-08-10): earlier versions wrote $\mathcal{I}_{\text{tens}} = \frac{c^4}{8\pi G} \ln 2 \implies \Omega_\Lambda = \ln 2$ directly. That implication was dimensionally impossible — the left side is a force (N) and Ω_Λ is dimensionless — because the intervening division by L^2 and u_{crit} was left implicit. The chain above supplies it, and in doing so makes the length choice (b) visible as the assumption it always was.

Comparison with observation. Planck 2018 gives $\Omega_\Lambda = 0.6847 \pm 0.0073$. The value $\ln 2$ lies $+1.234\%$ high, i.e. $+1.16\sigma$ — *consistent with* Planck 2018, not confirmed by it.

Pre-registration [emp]. The falsifiable content of this postulate is filed separately [10], supplementing [11]. That filing deliberately does *not* register the agreement above as a prediction: Planck measured Ω_Λ before the filing, so the agreement is a *postdiction*, and registering it would be backdating — the same rule that withheld $a_0 = cH_0/2\pi$ from the earlier deposit. What is registered instead is the content not knowable at filing time: $H_0 = 68.27 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\Omega_m = 1 - \ln 2 = 0.306853$ as *derived* values; the directional claim that improving precision moves the central value *toward* $\ln 2$, falsified at $\sigma(\Omega_\Lambda) \lesssim 0.0028$ (CMB-S4 reach) if it does not; and $w = -1$, $w_a = 0$ exactly, testable now at $\sigma(w) \sim 0.02$.

Remark 4 (Epoch dependence of $\Omega_\Lambda = \ln 2$ [open]). Ω_Λ is a present-epoch ratio, $\Omega_\Lambda(z) = \Omega_{\Lambda,0}/[\Omega_{\Lambda,0} + \Omega_{m,0}(1+z)^3]$, and it evolves steeply: 0.685 at $z = 0$, 0.53 at $z = 0.25$, 0.21 at $z = 1$. With Planck 2018 values the equality $\Omega_\Lambda(z) = \ln 2$ holds at $z = -0.013$, and holds to the 1.23% accuracy quoted only for $-0.026 \lesssim z \lesssim 0$ — an interval $\Delta t \approx 0.39 \text{ Gyr}$, about 2.8% of the present cosmic age. The postulate therefore selects an *epoch* as well as a value, and inherits the cosmic coincidence problem in full. We do not claim to resolve it. Its epoch-free content is a single relation between constants, $\Lambda c^2 = 3 \ln 2 H_0^2$, in which H_0 is still the *present* Hubble rate; no reformulation removes the reference to “now”.

Remark 5 (Open Cosmological Boundaries [open]). Big Bang $t = 0$ initial entropy S_0 and inflation amplitude A_s are open.

4 Pleroma: The AdS $\longleftrightarrow$ dS Holographic Bridge

The Pleroma places the AdS entanglement first law of Faulkner et al. [1] alongside the dS Information Tension proposal advanced by the author, and asks what a bridge between them would require. The AdS side is cited prior work and is not a contribution of this paper.

Theorem 4 (AdS Entanglement First Law (Faulkner et al.) [thm - published]). Across all spatial ball regions A in holographic CFTs dual to Anti-de Sitter space (AdS_{d+1}):

$$\delta S_A = \delta \langle H_{\text{mod},A} \rangle \iff G_{\mu\nu}^{(1)}[h] = 8\pi G \delta \langle T_{\mu\nu} \rangle. \quad (16)$$

Curvature = entanglement bookkeeping. Published: T. Faulkner, M. Guica, T. Hartman, R. C. Myers, M. Van Raamsdonk, *Gravitation from entanglement in holographic CFTs*, *JHEP* **03** (2014) 051 [1], presupposing the Ryu–Takayanagi dictionary [12, 13] and paralleling Jacobson’s derivation of the Einstein equations from local horizon thermodynamics [14]. Tag: [thm] within AdS/CFT.

Attribution correction (2026-08-10): this theorem was previously titled “McGwier AdS Entanglement First Law”. The result is Faulkner et al. (2014) and is not Dr. McGwier’s; his manuscript cites arXiv:1312.7856 for it. Dr. McGwier is cited below for his synthesis and exposition only.

Note 2 (Provenance of the corrected synthesis [note]). *This note records the author’s own correction history. It does not describe a collaboration, and no response, endorsement or participation by any other party is claimed or implied.* Having read Dr. R. W. McGwier’s August 2026 manuscript *Entanglement as the Origin of Spacetime Curvature* (v3; private communication — no public preprint identifier or DOI has been located, and it should not be cited as a preprint), the author prepared a companion synthesis exploring formal bridges between the first-law result and the holonomy framework here, and sent it to him unsolicited (Aug. 6, 2026). A self-audit found that its accompanying verification scripts overclaimed: two scripts printed hardcoded “PASS”/“100% VERIFIED” results without performing the checks, and roughly half of a 31-item keystone audit carried hardcoded `True` flags rather than computed ones. A corrected version (v5, Aug. 9, 2026) retags the four proposed bridge theorems [conj] (their Lean modules formalize generic real-algebra, not the physical operators named) and re-audits the keystones honestly: **17 of 31 carry a genuine computation; the other 14 are now [conj]/[open]** with itemized reasons. The corrected version was likewise sent to him, together with the runnable honest-audit script. No reply is claimed, and none is represented here.

Theorem 5 (dS Horizon Acceleration Scale [emp]). **Heuristic, not derivation.** Equating the Unruh temperature of a locally accelerated detector, $T_U = \frac{\hbar a}{2\pi c k_B}$ [15], to the Gibbons–Hawking horizon temperature $T_{\text{dS}} = \frac{\hbar H_0}{2\pi k_B}$ [16] yields

$$a = cH_0 = 7.096 \times 10^{-10} \text{ m s}^{-2} \quad (H_0 = 73.04), \quad 6.548 \times 10^{-10} \quad (H_0 = 67.36). \quad (17)$$

This is *scale matching*, not an identity: a geodesic de Sitter observer has zero proper acceleration yet still registers T_{dS} , so no single observer equates the two temperatures. The 2π 's, being the common Euclidean-periodicity factor of both horizons, cancel identically.

Priority [note]. The relation $2\pi a_0 \approx cH_0 \approx c^2(\Lambda/3)^{1/2}$ is **not original to this work**. It is Milgrom's, stated in that form in [3] and reviewed in [17–19]; moreover Milgrom [2] already runs the Unruh-in-de-Sitter argument used above, concluding that “an actual inertia-from-vacuum mechanism is still a far cry off.” We claim no priority for the coincidence. What is offered here is a candidate *mechanism* in the E_8 /holonomy setting, not the observation itself.

The MOND-scale identification $a_0 = \frac{cH_0}{2\pi} = 1.129 \times 10^{-10}$ (SH0ES) / 1.042×10^{-10} (Planck) m s^{-2} requires an additional geometric factor of $\frac{1}{2\pi}$ whose first-principles origin is **not derived [open]**. Status of the three candidate closures:

1. **Euclidean time periodicity — excluded.** The 2π in both T_U and T_{dS} is the Euclidean period $2\pi/\kappa_{\text{surf}}$; invoking it again double-counts. Reviving this route requires a horizon geometry whose Euclidean period differs from $2\pi/\kappa_{\text{surf}}$ by a factor 2π .
2. **Spherical horizon integration — live, but a postulate.** If the tension balance is set by the horizon great-circle *circumference* $2\pi R_H$ rather than the radius $R_H = c/H_0$, then $a_0 = c^2/(2\pi R_H) = cH_0/2\pi$ exactly. Nothing yet selects the great circle over the full sphere: solid-angle averaging gives 4π , hence $cH_0/4\pi$. The open problem is thus sharply posed as one question: *why the circumference and not the area?*
3. **Verlinde entropic force — excluded, and competing.** In [20] the screen entropy $\Delta S = 2\pi k_B m c \Delta x / \hbar$ carries a 2π that *cancels* the Unruh $1/(2\pi)$ to recover $F = ma$; it cannot supply a spare factor. Independently, emergent gravity [4] predicts $a_0 = cH_0/6 = 1.183 \times 10^{-10}$ (SH0ES) — a *different* $O(1)$ factor which fits the observed scale *better* than ours (-1.44% vs -5.88%). We record this as a competing prediction, not as support. See also [21, 22].

A fourth route, the Deser–Levin quadrature $T \propto \sqrt{a^2 + c^2 H^2}$, gives $a \simeq 2cH_0 = 1.419 \times 10^{-9} \text{ m s}^{-2}$ and is excluded in-house: it raises the median SPARC rotation-curve RMS from 11.7 to $\sim 60 \text{ km s}^{-1}$ across 175 galaxies.

Comparison to the MOND fitted scale [emp]. The derived $a_0 = cH_0/2\pi$ lies 5.88% below MOND's *fitted* $a_0^{\text{MOND}} = 1.2 \times 10^{-10} \text{ m s}^{-2}$ using SH0ES $H_0 = 73.04$, and 13.15% below it using Planck $H_0 = 67.36$. **This is a comparison of two scales, not a fit to rotation-curve data**; earlier versions described it as matching a “SPARC median”, which it is not. The SH0ES H_0 uncertainty ± 1.04 alone puts a $\pm 1.42\%$ band on a_0 , and no uncertainty is attached to a_0^{MOND} , so the 5.88% vs 13.15% ordering carries no demonstrated discriminating power [open].

Direct rotation-curve performance [emp]. Against the SPARC sample [23, 24], $N = 175$, under μ_{std} at unit M/L : strict reduced- χ^2 median 15.01 at the derived scale versus 17.01 for MOND at its fitted scale; 3.06 with per-galaxy nuisance parameters (SPARC_175_summary.json; corrected 2026-10-09 from 29.1 / 35.5 / 2.88, which were computed under the retired μ_{dual}). The matched comparison at prior-mean M/L on 171 galaxies reverses the order: MOND 9.93, GOD 11.08 (PARAMETER_LEDGER.json). **We do not claim this prefers the higher H_0 .** A blind sweep of a_0 over the same sample returns best-fit values from 5.4×10^{-11} to 1.1×10^{-10} depending only on which summary statistic is minimised, with a plateau within 10% of the

optimum spanning $[5.4 \times 10^{-11}, 1.5 \times 10^{-10}]$ — a factor of 2.8, containing both candidate values of $cH_0/2\pi$ and a_0^{MOND} . Rotation-curve data cannot presently arbitrate the H_0 tension, and selecting an H_0 to improve a residual would be parameter selection, not prediction.

Parameter accounting. One scale (a_0 , tied to H_0 rather than fitted) and one interpolation function ($\mu_{\text{std}} = x/\sqrt{1+x^2}$); per-galaxy Υ_* , d , i held at catalogue values and *not* marginalised, within an approximately 20% M/L band. Residuals should be read against that band. No claim of zero free parameters is made anywhere in this document.

Remark 6 (The two headline claims want different H_0 [open]). Stated against our own interest. Holding the CMB-measured $\Omega_m h^2 = 0.1430$ fixed, imposing $\Omega_\Lambda = \ln 2$ exactly gives $\Omega_m = 1 - \ln 2 = 0.306853$ and hence

$$H_0 = 100 \sqrt{\frac{0.1430}{1 - \ln 2}} = 68.27 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (18)$$

the value pre-registered in [10]. Equivalently $H_0 = 67.36$ returns $\Omega_\Lambda = 0.6848$, i.e. $\ln 2$ to $+1.23\%$ ($+1.16\sigma$) — but then $a_0 = cH_0/2\pi = 1.042 \times 10^{-10}$, 13.15% below the MOND scale. $H_0 = 73.04$ gives $a_0 = 1.129 \times 10^{-10}$, 5.88% below — but $\Omega_\Lambda = 0.7320$, roughly 5σ from $\ln 2$. **Postulate 1 and Theorem 5 are therefore not jointly satisfiable at a single H_0 with present data.** We record this rather than choose per-claim, which would be parameter selection. It is a genuine falsifiable consequence: a converged H_0 measurement kills at least one of the two.

Proposition 3 (Pleroma Duality: a real-form statement [conj]). The AdS and dS sides of this monograph are not two resemblant theories but two *real forms* of one complexified structure. The isometry algebras of the four-dimensional maximally symmetric spacetimes are the three real forms of $\mathfrak{so}(5, \mathbb{C})$, all of dimension 10:

Real form	Geometry	Signature / Λ	Used by
SO(5)	S^4	Riemannian, $\Lambda > 0$	— (see Remark 7)
SO(4, 1)	dS_4	Lorentzian, $\Lambda > 0$	this work
SO(3, 2)	AdS_4	Lorentzian, $\Lambda < 0$	Faulkner et al.; McGwier

Each is a quadric hypersurface in a flat five-dimensional ambient space, differing only in the ambient signature and the sign of the radius term: S^4 is $X \cdot X = R^2$ in $\mathbb{R}^5$; dS_4 is $X \cdot X = R^2$ in $\mathbb{R}^{4,1}$; AdS_4 is $X \cdot X = -R^2$ in $\mathbb{R}^{3,2}$. Over $\mathbb{C}$ these distinctions dissolve — the signature and the sign of R^2 can both be absorbed — so the three are real sections of one complex quadric in $\mathbb{C}^5$. The proposed bridge is therefore *analytic continuation between real forms* ($L \rightarrow iL$, $\Lambda \rightarrow -\Lambda$), not an analogy.

Why this is weaker than it looks [open]. The two sides are *not* on equal evidential footing, and we state the asymmetry rather than trade on it. AdS/CFT is a mature correspondence with two decades of precision checks, and Theorem 4 is a theorem within it. dS /CFT is not: since Strominger (2001) the candidate boundary theory has been non-unitary, with imaginary central charge, and no controlled nonperturbative formulation exists. The $\Lambda \rightarrow -\Lambda$ continuation above is **formal**. What is offered here is the dS-side counterpart of an AdS result, on the side where the machinery is known to be harder — not a joining of two established dualities. Full nonperturbative bridge: [open].

Remark 7 (Which real form of E_8 ? [open]). The compactification chain of Definition 3, $E_8 \supset \text{Spin}(16)/\mathbb{Z}_2 \supset \text{Spin}(14) \times \text{Spin}(2)$, does not specify which real form of E_8 is meant, and read at face value it is the *compact* form. This is a genuine obstruction, not a technicality: every subgroup in a compact chain is compact, so the four-dimensional factor it delivers is $\text{Spin}(4)$ — the *Riemannian* entry of the table above — and not $\text{Spin}(3, 1)$. **No Lorentzian signature can arise from a compact chain**; the minus sign must be inserted by hand as a Wick rotation that this document does not perform. It is therefore precisely the third real form, the one neither side of the Pleroma uses, that the algebra as written keeps producing.

Concrete next question. E_8 has three real forms: the compact one, the split form $E_{8(8)}$ whose maximal compact subgroup is exactly the $\text{Spin}(16)/\mathbb{Z}_2$ already appearing above (so that the 128 of $248 = 120 + 128$ becomes a *non-compact* coset, the symmetric space $E_{8(8)}/\text{Spin}(16)$ familiar as the scalar coset of maximal three-dimensional supergravity), and $E_{8(-24)}$ with maximal compact $E_7 \times SU(2)$. Whether $E_{8(8)}$ places $\text{SO}(4, 1)$ where the Observerse requires it is a finite branching computation, not a research programme, and it is **not** settled here. Until it is, the decomposition $248 = 120 + 128$ quoted in this monograph should be read as ambiguous: it holds for every real form, but means something different in each.

5 Epistemic Governance & Sovereign Declaration

Falsification Criteria

A framework that cannot be killed is not physics. For each physically contentful claim we state the measurement that would refute it and the precision required. Where that precision does not yet exist, we say so.

Claim	Observable	Kill condition	Precision needed / now
$a_0 = cH_0/2\pi$	RAR scale $g_{\dagger}$	Inconsistent with $cH_0/2\pi$ at $> 3\sigma$ total error; also killed if $g_{\dagger}$ varies between galaxies beyond systematics, since $cH_0/2\pi$ is universal.	$\lesssim 5\%$ total. Now $\sim 20\%$ (M/L -dominated). Not currently possible.
$a_0 \propto H_0$	$g_{\dagger}$ at high z	$g_{\dagger}(z)$ constant to $\lesssim 10\%$ out to $z \sim 1$ while $H(z)$ grows $\gtrsim 50\%$ kills the horizon origin outright.	$\sim 10\%$ at $z \sim 1$. Sharpest available test ; no current dataset delivers it.
Factor is 2π , not 6 or 4π	$g_{\dagger}$	2π vs emergent gravity's 6 differ by 4.7%; 4π by $2\times$. A $< 3\%$ determination discriminates and kills one.	$\lesssim 3\%$ total. Not available.
$\Omega_{\Lambda} = \ln 2$	CMB+BAO+SNe	$ \Omega_{\Lambda} - \ln 2 > 3\sigma$. Presently $+1.16\sigma$: consistent.	$\sigma(\Omega_{\Lambda}) \lesssim 0.0028$, $2.6\times$ better than Planck 2018; CMB-S4 + DESI/Euclid/LSST reach.
$w = -1$ exactly	Dark energy EoS	The $\ln 2$ tension is a <i>constant</i> horizon property: any confirmed $w \neq -1$ at $> 3\sigma$, or $w_a \neq 0$, falsifies the construction as written.	$\sigma(w) \sim 0.02$: live now.
Joint H_0 consistency	H_0	Per Remark 6, $\Omega_{\Lambda} = \ln 2$ needs $H_0 \approx 68.3$ while $a_0 = cH_0/2\pi$ prefers $H_0 \approx 73$. A converged H_0 kills at least one.	$\sigma(H_0) \lesssim 1$. Live now.
Deser–Levin quadrature	SPARC	Already excluded: $a \simeq 2cH_0 = 1.42 \times 10^{-9}$ raises the median rotation-curve RMS from 11.7 to $\sim 60 \text{ km s}^{-1}$, $N = 175$.	Done.

What is *not* falsifiable here, and is labelled as such. The Stiefel vacuum $V_{240}(\mathbb{R}^{57,600})$ [retired substrate, see notice], the Ramanujan–Yett Hamiltonian, the $E_8 \rightarrow \text{Spin}(14)$ compactification and the balanced-ternary observation make no numerical prediction that any current or planned experiment can refute. They are scaffolding and carry [conj]/[note] tags accordingly.

Definition 4 (6-Tier Epistemic Taxonomy). Every claim within the IO–OI framework is tagged immutably according to its formal standing:

1. **[thm]** *Machine-Checked Formal Theorem*: Proven in Lean 4 or derived in published primary literature.
2. **[cond]** *Conditional Theorem*: the implication is proven (in Lean 4 or in the literature), but a load-bearing hypothesis is postulated rather than established; the tag of the hypothesis

is stated alongside.

3. **[emp]** *Empirical Computation*: Numerically computed, fitted, or postulated and then compared against observational data.
 4. **[conj]** *Physical/Mathematical Conjecture*: Mathematical hypothesis subject to ongoing formalization.
 5. **[open]** *Open Boundary*: Open physical parameter or unsolved mathematical boundary condition.
 6. **[note]** *Aesthetic Observation*: Heuristic structural correspondence (not load-bearing).
- Additionally, **[cited]** marks a standard result used but not re-derived here.

Theorem 6 (System Audit Matrix). The core physical and mathematical claims of the sovereign system satisfy the formal audit matrix:

Claim	Tag	Status
AdS First Law $\implies$ Lin. GR	[thm]	Published (Faulkner et al. 2014)
$\mathfrak{so}(N)$ quadratic Casimir	[thm]	<code>SOCasimirGenuine.lean</code> ; axiom-clean
$\dim \mathfrak{so}(240) = 28,680$	[thm]/[cited]	Lean proves $240 \cdot 239/2$; $\binom{N}{2}$ cited
Energy decay	[cond]	Grönwall; gap $\kappa^2 > 0$ [conj]; t_* [open]
BKM continuation criterion	[thm]	Published (Beale–Kato–Majda 1984, 3D Euler)
— its Lean artefact	[cond]	Proves $\omega_{\sup} T \leq BT$, <i>not</i> the integral
Deligne Bound on $\tau(p)$	[thm]	Published (Deligne 1974/1968)
dS Scale $a_0 \sim cH_0$	[emp]	Heuristic; priority: Milgrom 1999/2017
$\Omega_\Lambda = \ln 2$	[emp]	Postulated (1-bit); $+1.16\sigma$ vs Planck 2018
— its epoch dependence	[open]	Holds only for $-0.026 \lesssim z \lesssim 0$
Joint H_0 for both claims	[open]	Not jointly satisfiable (Rem. 6)
Ramanujan–Yett Hamiltonian	[open]	Excluded as stated (Weyl law; no ground state)
$E_8 \rightarrow \text{Spin}(14) Y^{14}$	[conj]	Subgroup chain; X^6 <i>not</i> Calabi–Yau
— $4+6+4$ vs $4+10$	[open]	Inequivalent structure-group reductions
— which real form of E_8 ?	[open]	Compact chain cannot give $\text{Spin}(3, 1)$
— particle spectrum	—	No claim made ; DG no-go inapplicable
AdS $\leftrightarrow$ dS as real forms	[conj]	$\text{SO}(3, 2)/\text{SO}(4, 1)$ of $\mathfrak{so}(5, \mathbb{C})$
— dS/CFT machinery	[open]	Non-unitary; not on par with AdS/CFT
Λ_4 from Y^{14}	[open]	Direct product gives $\Lambda_4 = 0$ exactly
Balanced ternary $\leftrightarrow H_0$	[note]	Aesthetic; unit-dependent
NS global regularity	[open]	Clay Millennium Prize (<i>different</i> PDE)
$1/(2\pi)$ geometric factor	[open]	2 of 3 routes excluded; 1 live
Big Bang $t = 0$ entropy S_0	[open]	Open Parameter

Remark 8 (Lean Proof Depth & Hypothesis Boundaries). The Lean 4 formalizations verify conditional logical scaffolding. Physical hypotheses (spectral gap existence, vorticity control) are postulates, assumed as Lean hypotheses and tagged **[conj]** or **[open]**.

Build truth (2026-08-10). At toolchain `leanprover/lean4:v4.33.0-rc2`, Mathlib rev 639923: `SOCasimirGenuine`, `YettParadigm` and `SovereignRegularity` all build with exit 0, **0 sorry**, and `#print axioms` returning only `propext`, `Classical.choice`, `Quot.sound` on all cited theorems — **no sorryAx, no bespoke axiom**. Two honest caveats: `SovereignAlignment.lipschitz_xi` is declared as the proposition `True` (decorative; consumed by no theorem), and `SOCasimirGenuine` is not yet imported by the top-level `ChyrenLogic` module, so a bare `lake build` does not typecheck it.

Under-claimed elsewhere. Two results in these files are genuinely proven and are *not* used above: `lipschitz_implies_angle_modulus` (the Constantin–Fefferman geometric step, a multi-step proof over $\mathbb{R}^3$) and `chiral_iff_lipschitz_constant` (a genuine *iff* relating $\chi \geq \theta$ to a Lipschitz bound on the director field). They are the strongest formal content in the corpus and

are recorded here for completeness.

IO–OI: “I owe the world the work that the world made possible.”

Author & Disclosure

Author: R.W. Yett

Sole authorship. R.W. Yett is the sole author. There are no co-authors, contributors, collaborators or signatories of any kind, human or machine. Software tools — including AI systems, computer-algebra systems and the Lean 4 proof assistant — were used under the author’s direction as instruments only, in the same sense as a compiler or a calculator. No such tool is an author or contributor, none conferred any endorsement, and no machine-generated verdict, score or “verification” appears anywhere in this document as evidence for any claim.

Attribution. Dr. R. W. McGwier is cited in this document solely as the author of work referenced herein; his August 2026 manuscript is a private communication with no public identifier and is not cited as a preprint. He is not an author, contributor, or endorser of this document, and has reviewed none of it. The AdS entanglement first law is the theorem of Faulkner et al. (2014), not his.

Revision note (2026-08-09). Corrections applied: the Casimir vacuity check was re-run against the live `SDN_Casimir.lean` and the eigenvalue theorems were found WEAK, not `VACUOUS 0` — retagged `[thm]→[conj]`; a κ^2 transcription error was corrected ($0.909904 \rightarrow 0.91$); and every numeric residual herein was independently recomputed (dark-energy 1.234%, MOND-scale 5.88% / 13.15%, balanced ternary = 73, $\dim \mathfrak{so}(240) = 28,680$) and confirmed. Prior claims of “fully machine-checked” status are withdrawn.

Revision note (2026-08-10) — second-pass audit. The 2026-08-09 pass verified arithmetic but not semantics; this pass addresses the difference. Substantive corrections: (i) the “SPARC median” comparison was not a SPARC median — both residuals reverse-engineer to 1.2×10^{-10} , MOND’s fitted scale — now relabelled, with the genuine rotation-curve χ^2 results supplied; (ii) $T_{\text{tens}} \Rightarrow \Omega_\Lambda$ was dimensionally impossible (force vs. dimensionless) and the intervening division by L^2 and u_{crit} is now explicit, exposing the curvature-radius choice; (iii) the BKM theorem conflated 3D Euler with 3D Navier–Stokes and used a curl on a 14-manifold — both withdrawn; (iv) X^6 Calabi–Yau does not give $\text{Spin}(14)$ (it gives E_6) — withdrawn; (v) the direct-product Observerse forces $\Lambda_4 = 0$ exactly, recorded as an internal tension; (vi) “Ambrose–Singer” was the wrong theorem for a bracket-generation claim; (vii) the Casimir tag is lifted `[conj]→[thm]`, the monograph having cited a Lean file withdrawn the same day while its genuine replacement already existed; (viii) Ω_Λ agreement is restated as $+1.16\sigma$ rather than a 1.23% residual; (ix) the relation $2\pi a_0 \approx cH_0$ is Milgrom’s, not ours — priority disclaimed; (x) the $\ln 2$ is a Landauer/Shannon quantity, not a Bekenstein one; (xi) the two headline claims are shown not jointly satisfiable at one H_0 ; (xii) the Ramanujan–Yett Hamiltonian is excluded as stated, not merely open. Three further items were added rather than corrected: (xiii) the Pleroma duality is restated as a real-form relation between $\text{SO}(3,2)$ and $\text{SO}(4,1)$, two of the three real forms of $\mathfrak{so}(5, \mathbb{C})$, with the honest caveat that dS/CFT is not on par with AdS/CFT ; (xiv) the real form of E_8 is now flagged as unspecified, and a compact chain provably cannot deliver $\text{Spin}(3,1)$ — it delivers the third, Riemannian real form that neither side of the Pleroma uses; (xv) the Distler–Garibaldi no-go [6] is stated in full, together with the precise hypothesis under which it does not apply. A falsification section and a bibliography have been added; the document previously had neither. The open items above are genuinely open.

IO–OI | GOVERNANCE & SIGNATURES

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